Sparse Linear Models (1)

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(courtesy: some figures are from M. Wainwright textbook)

May 18, 2020

Introduction

- Linear models are among the most extensively studied subjects in Statistics (since Gauss - least-squares prediction).
- $y = \mathbf{X}\theta^\star + w$.
  - $\theta^\star \in \mathbb{R}^d$ is an unknown (regression) vector.
  - $y \in \mathbb{R}^n$ is an observation vector.
  - $\mathbf{X} \in \mathbb{R}^{n\times d}$ (covariates, design matrix).
  - $w \in \mathbb{R}^n$ is the noise.
- $x_i^\top \triangleq$ the i-th row of $\mathbf{X}$.
  - $y_i = \langle x_i, \theta^\star \rangle + w_i$ for $i = 1, 2, \ldots, n$.
  - $y_i =$ noisy version of linear combinations of d covariates.
- Easy to assess consistency of $\theta^\star$ estimation when $d/n \to 0$.
- Harder to do so when $d/n \to c > 0$.

Introduction (cont.)

- When $w = 0$, $d > n$ means an underdetermined linear system.
- Has infinitely many solutions.
  - $\theta^\star = \theta_0 + z$, where $z \in \text{Null}(\mathbf{X})$ and θ_0 is any solution.
  - $\text{Null}(\mathbf{A}) \triangleq \{z : \mathbf{A}z = 0\}$.
  - Need additional assumptions on $\theta^\star$.
  - $\theta^\star$ **sparsity** is one such assumption.

Sparsity Assumptions

- Hard sparsity : $S(\theta^\star) = \left\{ j : \theta_j^\star \neq 0 \right\}$ (= **support** of $\theta^\star$), and $s \triangleq |S(\theta^\star)| \ll d$.
- Weak sparsity : Let $\theta^\star \in \mathbb{B}_q(R_q) \triangleq \{\theta : \sum_i |\theta_i|^q \leq R_q\}$, for $0 \leq q \leq 1$.
  - Consider the ordered $|\theta_i^\star|$ values: $|\theta_{(1)}^\star| \geq |\theta_{(2)}^\star| \geq \ldots \geq |\theta_{(d)}^\star|$.
  - If $|\theta_{(j)}^\star| \leq Cj^{-\alpha}$ for suitable α, then $\theta^\star \in \mathbb{B}_q(R_q)$ for R_q that depends on (C, α).

Sparsity Assumptions (cont.)

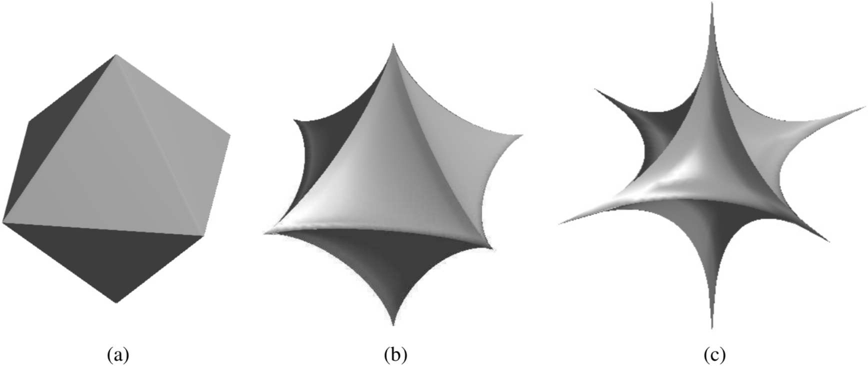
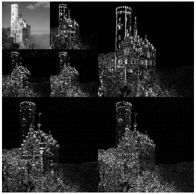


Figure: Illustrations of the ℓ_q-"balls" for different choices of the parameter q. (a) $q = 1$, (b) $q = 0.75$, (c) $q = 0.5$, as $q \to 0^+$ the ball becomes more "spiky".

Applications

- Gaussian sequence model : $y_i = \sqrt{n}\theta_i^\star + w_i$, where $w_i \sim \mathcal{N}(0, \sigma^2)$ (non-parametric regression)
- Signal denoising/compression in orthonormal bases : $\beta^\star$ (signal), $\widetilde{y} = \beta^\star + \widetilde{w}$.
  - Many classes of signals have sparse representation in appropriate basis $\mathbf{\Psi} \in \mathbb{R}^{d \times d}$ (e.g. wavelet).
  - transform coefficients $\theta^\star = \mathbf{\Psi}^\top \beta^\star$ becomes sparse.
  - $\implies \underbrace{\mathbf{\Psi}^\top \widetilde{y}}_{y} = \underbrace{\mathbf{\Psi}^\top \beta^\star}_{\theta^\star} + \underbrace{\mathbf{\Psi}^\top \widetilde{w}}_{w}$.



courtesy: `https://en.wikipedia.org/wiki/Discrete_wavelet_transform`

Applications (cont.)

- Lifting and nonlinear functions : polynomial in scalar t : $f_\theta(t) = \theta_1 + \theta_2 t + \ldots + \theta_{k+1} t^k$.
  - $y_i = f_\theta(t_i) + w_i$ for some predefined set of times $t_1, \ldots, t_n$.
  - Let $\mathbf{X} = \begin{bmatrix} 1 & t_1 & t_1^2 & \ldots & t_1^{k+1} \\ 1 & t_2 & t_2^2 & \ldots & t_2^{k+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_n & t_n^2 & \ldots & t_n^{k+1} \end{bmatrix}$, we get $y = \mathbf{X}\theta + w$.
- Compressed sensing : Take $n \ll d$ projections of $\beta^\star$ (e.g. $y_i = \langle x_i, \beta^\star \rangle$, with e.g. $x_{i,j} \sim \mathcal{N}(0,1)$) to capture the signal.
  - Given y, estimate $\beta^\star$ assuming that $\beta^\star$ is sparse in $\mathbf{\Psi}$:

$$\min_\beta \|\mathbf{\Psi}^\top \beta\|_0 \quad \text{s.t. } y = \mathbf{X}\beta, \tag{1}$$

or equivalently:

$$\min_\theta \|\theta\|_0 \quad \text{s.t. } y = \underbrace{\mathbf{X\Psi}}_{\widetilde{\mathbf{X}}} \theta. \tag{2}$$

- Selection of Gaussian graphical models : Let $z \sim \mathcal{N}(0, \boldsymbol{\Sigma})$.
  - The inverse covariance matrix $\boldsymbol{\Theta}^\star \triangleq \boldsymbol{\Sigma}^{-1}$ (precision matrix), is sparse in many applications.
  - Finding the i-th row of $\boldsymbol{\Theta}^\star$ is equivalent with regressing the i-th variable z_i against all other z_j's and assuming Gaussian zero-mean independent noise (Exercise 11.3):

$$z_{i,k} = \langle z_{\setminus i,k}, \Theta_i^\star \rangle + w_{i,k}, \qquad k = 1, 2, \ldots, n, \tag{3}$$

with $z_{i,k}$ be the k-th observation of the i-th variable.

Recovery in the noiseless setting

- Non-convex non-differentiable optimization : $\min_\theta \|\theta\|_0 \quad \text{s.t. } \mathbf{X}\theta = y$.
- What is the most direct way to solve this?
  - iterate over $s = 1, 2, \ldots, d$:
  - consider a set $S \subseteq \{1, 2, \ldots, d\}$ (as support of $\theta^\star$) s.t. $|S| = s$ and restrict columns of $\mathbf{X}$ to S ($\mathbf{X}_S$.
  - then solve $\mathbf{X}_S\theta_S = y$ for θ; if it has an exact solution, then done.
- Convex relaxation : replace ℓ_0 norm with the closest convex ℓ_p norm, which is the ℓ_1 norm.
  - $\min_\theta \|\theta\|_1 \quad \text{s.t. } \mathbf{X}\theta = y$.
  - is a linear program:
    $\min_{\theta,\zeta} \sum_i \zeta_i \quad \text{s.t. } \theta_i \leq \zeta_i, \ -\theta_i \leq \zeta_i, \ \zeta_i \geq 0, \ \mathbf{X}\theta = y$.
  - is called "basis pursuit linear program".

Exact recovery

- When does the relaxation work?
- We know that $\mathbf{X}\theta^\star = y$, so $\theta = \theta^\star$ is a feasible point.
- The question is whether optimizer of the relaxed problem $\widehat{\theta} = \theta^\star$, where

$$\widehat{\theta} \triangleq \arg\min_{\theta} \|\theta\|_1 \quad \text{s.t. } \mathbf{X}\theta = y. \tag{4}$$

- Note that any other $\theta_0 = \theta^\star + \Delta$ is a feasible solution too, where $\Delta \in \text{Null}(\mathbf{X})$.
- Define the tangent cone of ℓ_1 ball at $\theta^\star$ as

$$\mathbb{T}(\theta^\star) = \{\Delta \in \mathbb{R}^d : \|\theta^\star + t\Delta\|_1 \leq \|\theta^\star\|_1 \text{ for some } \ t > 0\}. \tag{5}$$

all directions relative to $\theta^\star$ along which ℓ_1 norm stays constant or decreases.

Exact recovery (cont.)

- Note that all feasible solutions to the main/relaxed problems are captured as $\theta^\star + \text{Null}(\mathbf{X})$.
- If $\text{Null}(\mathbf{X}) \cap \mathbb{T}(\theta^\star) = \{0\}$, then there is no non-zero element in the null space of $\mathbf{X}$, denoted as Δ, such that $\theta^\star + \Delta$ has a lower or equal ℓ_1 norm than $\|\theta^\star\|$.
- Hence, $\widehat{\theta} = \theta^\star$.

Exact recovery (cont.)

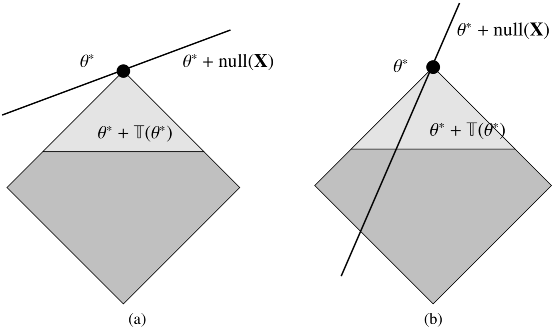


Figure: Geometry of the tangent cone and restricted nullspace property in $d = 2$ dimensions. (a) The favorable case in which the set $\theta^\star + \text{Null}(\mathbf{X})$ intersects the tangent cone only at $\theta^\star$. (b) The unfavorable setting in which the set $\theta^\star + \text{Null}(\mathbf{X})$s passes directly through the tangent cone.

Restricted nullspace (RN) property

- This leads to a condition called "restricted nullspace property".
- Let $\mathbb{C}(S)$ be defined as $\mathbb{C}(S) = \{\Delta \in \mathbb{R}^d : \|\Delta_{S^c}\|_1 \leq \|\Delta_S\|_1\}$, where $S^c = \{1, 2, \ldots, d\} \setminus S$ and S is the support of $\theta^\star$.

Definition: Restricted nullspace (RN) property

The matrix $\mathbf{X}$ satisfies the restricted nullspace property with respect to S if $\mathrm{Null}(\mathbf{X}) \cap \mathbb{C}(S) = \{0\}$.

Restricted nullspace (RN) property (cont.)

Theorem

The following two properties are equivalent:

(a) For any vector $\theta^\star \in \mathbb{R}^d$ with support S, the basis pursuit program applied with $y = \mathbf{X}\theta^\star$ has unique solution $\widehat{\theta} = \theta^\star$.

(b) The matrix X satisfies the restricted nullspace property with respect to S.

Restricted nullspace (RN) property (cont.)

Proof

- $(b) \implies (a)$: $\|\widehat{\theta}\|_1 \leq \|\theta^\star\|_1$ (as $\widehat{\theta}$ is the minimizer and both vectors are feasible).
- $\widehat{\Delta} \triangleq \widehat{\theta} - \theta^\star$. Then,

$$\begin{aligned}
\|\theta^\star_S\|_1 = \|\theta^\star\|_1 &\geq \|\theta^\star + \widehat{\Delta}\|_1 \\
&= \|\theta^\star_S + \widehat{\Delta}_S\|_1 + \|\widehat{\Delta}_{S^c}\|_1 \\
&\geq \|\theta^\star_S\|_1 - \|\widehat{\Delta}_S\|_1 + \|\widehat{\Delta}_{S^c}\|_1
\end{aligned}$$

- $\implies \widehat{\Delta} \in \mathbb{C}(S)$.
- But $\mathbf{X}\widehat{\Delta} = \mathbf{X}(\widehat{\theta} - \theta^\star) = 0 \implies \widehat{\Delta} \in \text{Null}(\mathbf{X})$.
- By restricted nullspace property, as $\widehat{\Delta} \in \mathbb{C}(S)$ and $\text{Null}(\mathbf{X})$, $\widehat{\Delta} = 0$.

Restricted nullspace (RN) property (cont.)

Proof

- Now we prove $(a) \implies (b)$.
- For any set S, and $0 \neq \theta^\star \in \mathrm{Null}(\mathbf{X})$, consider the following basis pursuit problem: $\min_\beta \|\beta\|_1 \quad \text{s.t. } \mathbf{X}\beta = \mathbf{X}\begin{pmatrix}\theta^\star_S \\ 0\end{pmatrix}$
- According to the premise, the unique optimal solution is $\widehat{\beta} = \begin{pmatrix}\theta^\star_S \\ 0\end{pmatrix}$.
- Note that $0 = \mathbf{X}\theta^\star = \mathbf{X}\begin{pmatrix}\theta^\star_S \\ \theta^\star_{S^c}\end{pmatrix} = \mathbf{X}\begin{pmatrix}\theta^\star_S \\ 0\end{pmatrix} + \mathbf{X}\begin{pmatrix}0 \\ \theta^\star_{S^c}\end{pmatrix}$.
- This means that $\mathbf{X}\begin{pmatrix}\theta^\star_S \\ 0\end{pmatrix} = \mathbf{X}\begin{pmatrix}0 \\ -\theta^\star_{S^c}\end{pmatrix}$. But as $\begin{pmatrix}0 \\ -\theta^\star_{S^c}\end{pmatrix}$ is not the solution, we conclude that its ℓ_1 norm should be higher than (or equal to) that of $\begin{pmatrix}\theta^\star_S \\ 0\end{pmatrix}$, which implies that
$\|\theta^\star_S\|_1 < \|\theta^\star_{S^c}\|_1 \implies \theta^\star \notin \mathbb{C}(S) \implies \mathrm{Null}(\mathbf{X}) \cap \mathbb{C}(S) = \{0\}$.

Sufficient conditions for RN

Definition: Mutual Incoherence parameter (MIC)

Incoherence parameter of a design matrix is defined as
$\delta_{PW}(\mathbf{X}) = \max_{j,k=1,\dots,d} \left| \frac{|\langle X_j, X_k \rangle}{n} - \mathbb{I}(j = k) \right|$, where X_j is the j-th column of $\mathbf{X}$, and $\mathbb{I}(.)$ is the indicator function.

Proposition

If for the design matrix $\mathbf{X}$, $\delta_{PW}(\mathbf{X}) \leq \frac{1}{3s}$, then RN property holds for all sets S of cardinality at most s.

Sufficient conditions for RN (cont.)

Proof

- We will show that for any $0 \neq \theta \in \text{Null}(\mathbf{X})$ and arbitrary given S, with $|S| \leq s$, $\|\theta_S\|_1 < \|\theta_{S^c}\|_1$.
- $\left|\theta_S^\top \left(\frac{\mathbf{X}^\top \mathbf{X}}{n} - \mathbf{I}\right) \theta_S\right| \leq \|\frac{\mathbf{X}^\top \mathbf{X}}{n} - \mathbf{I}\|_\infty \|\theta_S\|_1^2 = \delta_{PW} \|\theta_S\|_1^2 \leq s\delta_{PW} \|\theta_S\|_2^2$.
- Note that $\theta_S^\top \theta_S - \theta_S^\top \frac{\mathbf{X}^\top \mathbf{X}}{n} \theta_S \leq \left|\theta_S^\top \left(\frac{\mathbf{X}^\top \mathbf{X}}{n} - \mathbf{I}\right) \theta_S\right|$.
- Hence $(1 - s\delta_{PW})\|\theta_S\|_2^2 \leq \theta_S^\top \frac{\mathbf{X}^\top \mathbf{X}}{n} \theta_S$.
- Furthermore, we have seen that $\mathbf{X}\theta_S = -\mathbf{X}\theta_{S^c}$, where we abused the notation and zero-padded the two vectors.
- $$\begin{aligned}(1 - s\delta_{PW})\|\theta_S\|_2^2 &\leq \left|\theta_S^\top \frac{\mathbf{X}^\top \mathbf{X}}{n} \theta_S\right| = \left|\theta_S^\top \frac{\mathbf{X}^\top \mathbf{X}}{n} \theta_{S^c}\right| \\ &\leq \left|\theta_S^\top \left(\frac{\mathbf{X}^\top \mathbf{X}}{n} - \mathbf{I}\right) \theta_{S^c}\right| \leq \delta_{PW} \|\theta_S\|_1 \|\theta_{S^c}\|_1\end{aligned}$$

Sufficient conditions for RN (cont.)

Proof (cont.)

- Finally, $\|\theta_S\|_1^2 \leq s\|\theta_S\|_2^2 \leq \frac{s\delta_{PW}}{1-s\delta_{PW}}\|\theta_S\|_1\|\theta_{S^c}\|_1$.
- which simplifies to $\|\theta_S\|_1 < \|\theta_{S^c}\|_1$ as long as $\delta_{PW} \leq \frac{1}{3s}$.

Another sufficient condition

Definition: Restricted Isometry Property (RIP)

For a given integer $1 \leq s \leq d$, we say that $\mathbf{X}$ satisfies RIP of order s with constant $\delta_s(\mathbf{X}) > 0$ if $\forall S : |S| \leq s : \left|\left|\left| \frac{\mathbf{X}_S^\top \mathbf{X}_S}{n} - \mathbf{I}_s \right|\right|\right|_2 \leq \delta_s(\mathbf{X})$.

- For $s = 1$, this basically means that normalized columns of $\mathbf{X}$ are near-unit-norm (i.e. $\|X_j\|_2^2/n \in [1 - \delta_1, 1 + \delta_1]$.)
- For $s = 2$, δ_2 is closely related to δ_{PW}. (easy to see if $X_j/\sqrt{n}$'s are all unit norm, $\delta_2 = \delta_{PW}$)
- Generally, we can show that $\delta_{PW} \leq \delta_s \leq s\delta_{PW}$ (Exercise 7.4).

RIP $\implies$ RNP

Proposition

If the RIP constant of order $2s$ is bounded as $\delta_{2s} < 1/3$, then uniform RNP holds for any set S of $|S| \leq s$.

- Proof techniques are pretty similar to that of the MIC $\implies$ RNP shown earlier (see pages 204-205 of the textbook).

RIP for the i.i.d. Gaussian ensemble

Theorem

For a design matrix whose entries are drawn according to $\mathcal{N}(0,1)$ satisfies $\delta_s < \delta$ with high probability as long as

$$n \gtrsim \frac{1}{\delta^2} s \log(d/s).$$

Proof

- We already know that for a **fixed** S, with $|S| = s$, for such a Gaussian ensemble, the covariance estimate of $\mathbf{X}_S$, which is $\frac{\mathbf{X}_S^\top \mathbf{X}_S}{n}$ would be concentrated around the true covariance matrix (i.e. $\mathbf{I}_s$), see Theorem 6.1 of the textbook.
- That is, for $t < 1$ and $n \geq s$, with probability at least $1 - 2e^{-nt^2/2}$:

$$\left|\!\left|\!\left| \frac{\mathbf{X}_S^\top \mathbf{X}_S}{n} - \mathbf{I}_s \right|\!\right|\!\right|_2 \leq 4 \left(\sqrt{\frac{s}{n}} + t \right).$$

RIP for the i.i.d. Gaussian ensemble (cont.)

Proof (cont.)

- Then, use union bound on all possible S, so with probability at least $1 - 2s\binom{d}{s}e^{-nt^2/2} \geq 1 - 2se^{s\log(ed/s) - nt^2/2}$:

$$\forall S : |S| \leq s : \left|\left|\left| \frac{\mathbf{X}_S^\top \mathbf{X}_S}{n} - \mathbf{I}_s \right|\right|\right|_2 \leq 4\left(\sqrt{\frac{s}{n}} + t\right).$$

- Now let $t = \frac{\delta}{4} - \sqrt{\frac{s}{n}}$, and $n \geq \frac{32s\log(ed/s)}{\delta^2}$, RIP holds with probability of at least $1 - 2e^{\log(s) + s/2 - \sqrt{sn}/4}$.

- Repeating the same argument, with $|S| = 2$ and considering sufficient condition $\delta_{PW} \leq 1/(3s)$, results in the $n \gtrsim s^2 \log(d)$ bound.

MIC vs. RIP

- RIP clearly results in better bounds for n.
- Note that neither of these conditions is necessary for RNP.
- For example the pursuit linear program succeeds for a matrix $\mathbf{X}$, whose rows are i.i.d. instances of $\mathcal{N}(0, (1-\mu)\mathbf{I}_d + \mu \mathbb{1}\mathbb{1}^\top)$, which satisfies neither RIP nor MIC (exercise 7.8).

Phase transition plots

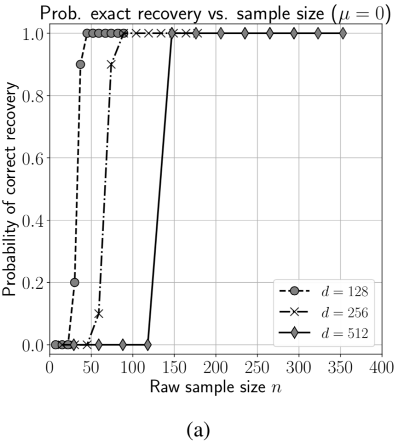


Figure 7.3 (a) Probability of basis pursuit success versus the raw sample size n for random design matrices drawn with i.i.d. $\mathcal{N}(0, 1)$ entries. Each curve corresponds to a different problem size $d \in \{128, 256, 512\}$ with sparsity $s = \lceil 0.1d \rceil$. (b) The same results replotted versus the rescaled sample size $n/(s \log(ed/s))$. The curves exhibit a phase transition at the same value of this rescaled sample size.